

Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Neonatal respiratory distress syndrome (NRDS)
- B. Tracheoesophageal fistula
- C. Pneumonia
- D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

<reasoning process here> The baby was born at 32 weeks gestation, which is preterm but not extremely so. The presence of tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis suggests respiratory distress. Oxygen saturation of 88% on room air indicates hypoxemia. The fact that cyanosis is responding to oxygen administration suggests that the underlying issue is respiratory in nature, not a cardiac shunt, which would typically present with persistent cyanosis despite oxygen.

Neonatal respiratory distress syndrome (NRDS) is common in preterm infants due to surfactant deficiency, which is particularly prevalent in those born before 34 weeks. The clinical presentation of tachypnea, retractions, nasal flaring, and hypoxia matches NRDS. The mother had gestational diabetes, which is a known risk factor for neonatal respiratory distress due to fetal lung immaturity and potential macrosomia, although it is not a direct cause of NRDS.

Tracheoesophageal fistula (B) would present with feeding difficulties, vomiting, and air in the stomach, not necessarily with cyanosis and respiratory distress. Pneumonia (C) may present with similar signs but would typically have a history of infection or fever, and the response to oxygen would be less predictable. A congenital heart anomaly with right-to-left shunt (D) would present with persistent cyanosis, even with oxygen, and would be more likely to have signs of poor perfusion or polycythemia.

The baby's symptoms are consistent with NRDS, especially in a preterm infant. The nasogastric tube placement without issue suggests no esophageal anomaly or obstruction.

Therefore, the most likely diagnosis is neonatal respiratory distress syndrome.

<answer> A. Neonatal respiratory distress syndrome (NRDS) </answer>